

GPU Power and Performance Prediction Challenge

1. Background

LLM inference deployment is one of the most critical engineering challenges in AI infrastructure. Every user request consumes GPU power and introduces latency — two factors that directly determine data center operating costs and user experience.

Imagine you are an energy efficiency architect at an AI cloud computing company. Your boss hands you a spreadsheet recording measured power draw and latency across various GPU models, LLM architectures, and request loads, then asks:

"Given hardware specs, model parameters, and expected traffic — can we predict how much power this machine will consume and how fast it will be?"

That is the problem you are here to solve.

This dataset originates from the academic paper *Watt Counts: An Empirical Study of GPU Power Consumption and Performance in LLM Inference*. It contains thousands of real measurements across 8 data-center-class GPUs, 30+ open-source LLMs, in both offline batch and online serving scenarios. Your task is to **simultaneously predict four core performance metrics** from three input dimensions: **hardware specs + model parameters + workload**.

2. Task Definition

Input (X) — the following features for each experimental run:

Dimension	Feature	Description
Workload	lambda_qps	Request arrival rate (queries/sec), -1 = offline batch
Workload	scenario	Operating scenario: offline / server_low / server_high
Workload	avg_prompt_tokens	Average input tokens per request
Workload	avg_generation_tokens	Average output tokens per request

Dimension	Feature	Description
Workload	num_requests	Number of requests in the experiment
Model	total_b_params	Total model parameters (billions)
Model	num_layers	Number of transformer layers
Model	hidden_size	Hidden layer dimension
Model	num_attention_heads	Number of attention heads
Model	num_key_value_heads	Number of KV heads (GQA)
Model	model_type	Model architecture family (e.g., llama, qwen2)
Hardware	memory_size_gb	GPU memory capacity (GB)
Hardware	release_year	GPU release year
Hardware	base_clock_mhz	GPU base clock (MHz)
Hardware	boost_clock_mhz	GPU boost clock (MHz)
Hardware	architecture	GPU architecture name
Hardware	memory_type	Memory type (HBM2e/GDDR6/etc.)

Output (y) — four continuous target variables:

Target	Meaning	Unit	Business Value
gpu_power_draw_watts	Average GPU power draw	Watts (W)	Electricity cost estimation
avg_e2e_latency_seconds	Average end-to-end latency	Seconds (s)	User experience
energy_efficiency_tokens_per_joule	Energy efficiency (tokens/joule)	Token/J	"Green AI" metric
throughput_tokens_per_second	System throughput (tokens/sec)	Token/s	Service capacity

This is a **multi-target regression** task. Build a model that predicts all four metrics simultaneously from hardware, model, and workload features.

3. Dataset

3.1 Data Source

The data comes from the Watt Counts Subset in the *Watt Counts* paper, measuring real inference performance of 30+ open-source LLMs (from GPT-2 124M to Gemma-3 27B) across 8 GPUs (from Tesla T4 to H200 NVL) in both offline batch and online serving scenarios.

3.2 File Description

File	Description
train.csv	Training set with feature columns + 4 target columns
A.csv	A-leaderboard test set, features only, NO targets
B.csv	B-leaderboard test set, features only, NO targets
baseline.ipynb	Contestant baseline notebook for getting started
Paper_Watt_Counts.pdf	The <i>Watt Counts</i> original paper

3.3 Data Characteristics

- Train, A-test, and B-test sets are randomly split at a 70:15:15 ratio.
- **Training set has GPU imbalance:** rare high-end GPUs (H200, H100, A100) are intentionally downsampled, simulating real-world data scarcity for expensive hardware. Common GPUs (T4, L4, L40S) have full representation.
- A and B test sets remain fully balanced across all GPU types.
- Values span extreme ranges — from a 0.5B model on a low-end GPU to a 27B model on an H200 at full throttle.
- Offline mode entries have `lambda_qps = -1` , representing continuous batch processing without request gaps.

4. Evaluation Metrics

This task uses a **composite score** combining the following metrics:

4.1 Primary Metric: R^2 (Coefficient of Determination)

For each target variable i :

$$R^2 * i = 1 - \frac{\sum_j (y * ij - \hat{y} * ij)^2}{\sum_j (y * ij - \bar{y}_i)^2}$$

The average R^2 across all four targets forms the primary score (range 0 ~ 1):

$$\text{Primary Score} = \frac{1}{4} \sum_i R_i^2$$

Higher is better (1.0 = perfect prediction).

4.2 Auxiliary Metric: WMAPE (Weighted Mean Absolute Percentage Error)

$$\text{WMAPE}_i = \frac{\sum_j |y * ij - \hat{y} * ij|}{\sum_j |y * ij|} \times 100\%$$

The average WMAPE across all four targets serves as auxiliary reference (lower is better).

5. Submission Format

Submit a CSV file named `A_predict.csv` (for A-leaderboard) or `B_predict.csv` (for B-leaderboard) with the following format:

```
gpu_power_draw_watts,avg_e2e_latency_seconds,energy_efficiency_tokens_per_joule,throughput_token
163.42,0.0487,27.68,4567.2
45.12,0.0125,18.34,892.1
...
```

- **Order must match the test set exactly** (first prediction row = first test set row)
- **Column names must match** the four target names above exactly
- **Do NOT include row indices or ID columns**

6. Constraints

To ensure fair competition:

Constraint	Limit
Pre-trained models	No pre-trained weights allowed (including those trained on non-public data)
LLM APIs	No LLM API calls (GPT-4, Claude, etc.)
Libraries	Only standard NOAI libraries allowed (numpy , pandas , scikit-learn , torch , xgboost , lightgbm , etc.)
External data	Public GPU/LLM specs are allowed, but any external copies or variants of this dataset are strictly forbidden

7. References

1. *Watt Counts: An Empirical Study of GPU Power Consumption and Performance in LLM Inference* — original paper for this dataset
2. *LLM Inference Performance Engineering* — NVIDIA Technical Blog series
3. *Roofline Model* — classic performance analysis framework

8. License

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